

The Utility Problem

Phase balance in an electricity grid refers to the balance of the electrical power flowing through the three phases of the grid (A, B, and C). In a balanced grid, the power flowing through each phase is equal. However, when there is an imbalance, it means that the power flowing through one or more of the phases is different from the others.

An imbalance in the phase distribution can cause several problems, such as:

- Reduced system efficiency and reliability - an imbalance in the phase distribution can cause an increase in system losses, reducing the overall efficiency and capacity of the grid, which can lead to outages or safety breaches.
- Voltage fluctuations - an imbalance in the phase distribution can cause voltage fluctuations, which can affect the performance of equipment connected to the grid.
- Harmonics - an imbalance in the phase distribution can cause harmonic distortion, which can affect the power quality of the grid.

In the past, without the use of data analytics, correcting phase imbalance would have typically involved upgrading the power lines with a higher diameter conductor or a higher capacity transformer, which would have been a costly and time-consuming solution. With the use of data analytics, it is possible to identify which phase of the grid is more or less loaded and then redistribute the load across the three phases to ensure that the power flowing through each phase is equal.

Using data analytics, utilities can monitor the grid in real-time and identify potential issues before they become critical. This allows them to take proactive measures to correct phase imbalance, such as load balancing, instead of waiting for an imbalance to occur and then taking reactive measures.

By identifying which phase is more or less loaded, utilities can also identify opportunities to optimize the grid by switching some loads from a heavily loaded phase to a less loaded one. This can help to reduce the overall load on the grid, improve system efficiency, and prevent equipment damage. Additionally, switching some loads from a heavily loaded phase to a less loaded one can be done with a relatively low investment cost compared to upgrading the conductors.

The Data Solutions and Methods

For this use case, it is possible to use the data from the "Phase Balance Analysis" use case to find the locations with the highest imbalance. This use case helps to correct these occurrences. To avoid having to manually find single-phase feeders suitable for reconnection, an automated algorithm is used to recommend single-phase feeders to reconnect for better phase balance. The North American grid is typically a three-phase feeder from the transformer, and then single-phase feeders are routed directly to the customers (feeder laterals). This single-phase

feeder may have a varying number of customers. There is, therefore, the potential for reconnecting these single-phase laterals to another phase.

From the time series data of the measurements on the transformer or feeder, it is determined how the single-phase load or loads should ideally look like, which will be switched from the most loaded phase to the other phases - for each time stamp, the average of all three phases is calculated, divided by three. This calculates the ideal load for each phase. This ideal load is then subtracted from the individual phases to find the deviations that need to be corrected. The load on each single-phase feeder is then determined, and the feeders which are on the most heavily loaded phase and whose loads are as close as possible to this ideal waveform are searched for. Finally, the single-phase laterals that most closely resemble this ideal load are selected for reconnection.

The data needed for this use case:

- Time series data measurements at the transformer or the feeder - the predefined function retrieves time series data with the selected granularity.
- Time series data of measurements at the consumption points (customers) - the predefined function retrieves the time series data with the selected granularity.
- Geographic Information System (GIS) data together with connectivity - predefined EDM function retrieves all connected meters on the feeder alongside the information about the meter's phase assignment.

The Awesense Energy Transition Platform will make all the necessary data available (data already cleaned, and synchronized to common timestamps), which can be retrieved very easily using predefined functions and API calls.

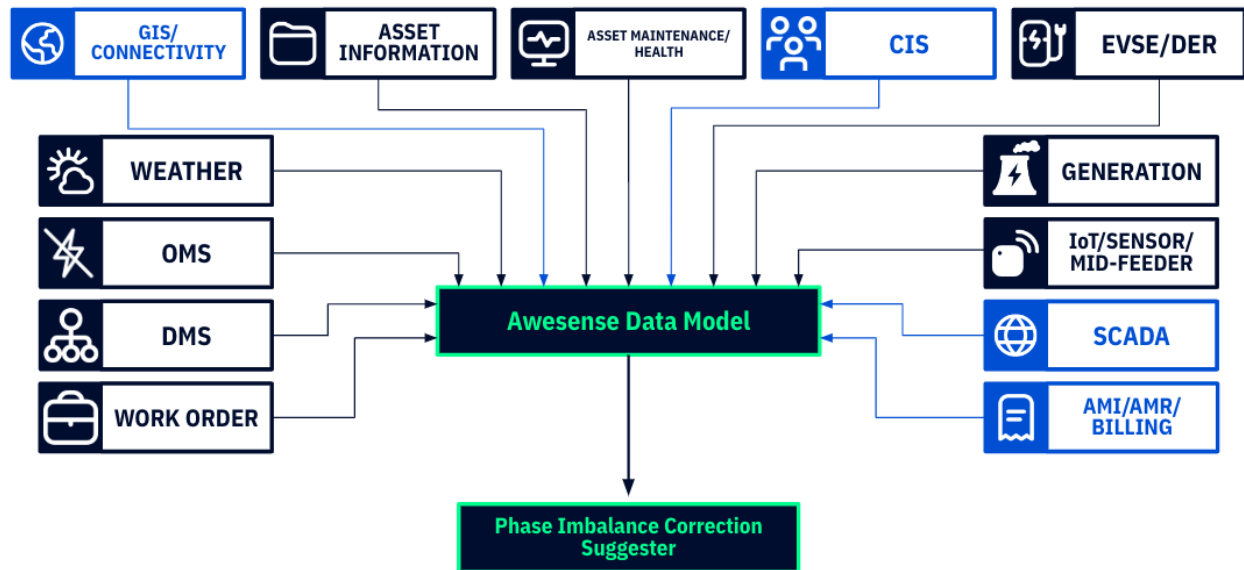
Phase Imbalance Correction Suggester

[UC40-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.